Challenge 1: The Virtual Patient Encounter System

Theme: Digital Twin Simulation • Healthcare Education • Generative AI • Human-Computer Interaction • Virtual Simulation

Challenge Summary

Healthcare students need repeated opportunities to practice clinical interviewing, communication, and clinical reasoning. Traditional simulations can be expensive, difficult to scale, and often unavailable when students need additional practice.

Your challenge is to build a reusable AI-powered virtual patient encounter system that creates realistic clinical encounters from structured patient profiles. The experience should go beyond a simple chatbot by allowing students to interact with a virtual patient while simultaneously accessing relevant patient medical information, similar to how clinicians interact with a patient while reviewing their chart.

The system should support the creation, management, and execution of multiple virtual patients without requiring patient-specific code. A faculty member should be able to create a new patient profile using your system and then immediately conduct an encounter with that patient.

Your Challenge

Your solution should consist of three core components:

1. Patient Studio — Faculty-Facing

Build a way for faculty to create and manage virtual patients.

The system should allow faculty to:

- Create a patient profile using a structured or descriptive approach.
- Add knowledge sources such as papers, guidelines, or institutional protocols.
- Attach clinical artifacts such as laboratory reports, ECGs, X-rays, CT scans, or prior notes.
- Define when clinical information and artifacts should become available during the encounter.
- Review and correct generated patient information.

- Check that the patient's clinical information is internally consistent.
- Approve and publish patient profiles for students.
- Maintain multiple reusable patient profiles.

2. Encounter Interface — Student-Facing

Students should be able to conduct a realistic clinical encounter with any published patient profile.

The interface must include a **split-screen experience**:

Left side:
Student interaction with the virtual patient.

Right side:
The patient's medical information and clinical record, dynamically displayed as relevant information becomes available.

The encounter should:

- Run any published patient profile without patient-specific code.
- Support text interaction, with voice and/or avatar interaction encouraged.
- Maintain a consistent patient identity and conversation state.
- Display relevant medical records, such as prior notes, structured patient information, and clinical artifacts.
- Release information progressively based on the patient's profile and the student's questions.
- Prevent the patient from inventing clinical facts outside the approved profile.
- Maintain a record of which information has been disclosed during the encounter.

3. Encounter Feedback — Student-Facing

After the encounter, the system should provide feedback based on the learning objectives defined for the patient.

Feedback should:

- Identify important findings the student elicited and missed.
- Evaluate the structure, sequencing, and quality of the interview.
- Evaluate communication, empathy, clarity, and professionalism.
- Provide specific and explainable feedback linked to the encounter.
- Identify both strengths and areas for improvement.
- Provide actionable suggestions for the student's next attempt.
- Be driven by faculty-defined learning objectives.
- Log each clinical trial/encounter.

Clinical Validity and Safety

The virtual patient should remain grounded in its approved patient profile and clinical content.

The system should:

- Maintain consistency between the patient's history, medications, diagnoses, laboratory results, and clinical artifacts.
- Prevent unsupported clinical information from being presented as fact.
- Use faculty-approved or validated clinical information when generating dynamic findings.
- Ensure that attached medical records remain consistent with the patient profile.

Base Level

A **Base-level solution must demonstrate the complete core experience**, including:

- Faculty patient-profile creation
- Patient knowledge and artifact management
- Profile review and publication
- Reusable patient profiles
- Complete student encounter
- Split-screen patient interaction + dynamic medical record
- Progressive information disclosure
- Patient memory/state
- Clinical grounding and consistency
- Encounter logging
- Post-encounter assessment and explainable feedback

Stretch

Teams that complete the Base experience may extend their solution with advanced capabilities such as:

- Real-time voice interaction
- Avatar-based patient interaction (ideal)
- Multimodal patient interaction
- Adaptive patient behavior
- Advanced memory/state architectures
- More sophisticated clinical artifact generation or interaction
- Integration APIs for external SaaS application integration (Canvas etc.)

Suggested Technologies

Teams are **not required to use any specific technology**.

Potential approaches include:

- Cloud-native GenAI applications
- LLMs
- Agentic AI
- RAG
- Multimodal models
- Speech recognition and synthesis
- NLP
- Streaming/real-time responses
- Voice and avatar systems
- Memory and state-management architectures
- Agent orchestration
- Contextual engineering
- Observability

Resources Provided

Teams will have access to:

- Synthetic patient personas
- Existing VHS prompts
- Sample clinical encounters and case materials
- Learning objectives
- Example documentation templates
- AWS Bedrock access and credits

Deliverables

Teams must submit:

1. Working prototype demonstrating patient-profile creation, a complete patient encounter, and generated feedback.
2. Code repository containing the implementation and infrastructure code, if applicable.
3. Short demonstration video or live demonstration.

4. Four-minute pitch explaining the solution, architecture, technical approach, and educational value.

Evaluation Considerations

Solutions will be evaluated on:

- Architecture and reusability
- Quality and clinical validity of patient profiles
- Technical innovation
- User experience
- Quality of the clinical encounter
- Effectiveness and explainability of feedback

Key Demonstration

During judging, a faculty member will create a new patient profile using the team's Patient Studio and then conduct an encounter with that newly created patient.

The solution should demonstrate that the newly created patient can immediately operate within the same reusable encounter architecture without patient-specific code changes.